accuracy of how well your device tracks objects placed in AR and how they would move in the environment.

Horizontal Plane and Light detection

ARKit allows your device to analyze the environment and detect horizontal planes for objects you might wish to place in AR. This means floors, tables, or any flat surfaces. ARKit can also use your camera sensor to detect lighting and thus apply the correct amount of lighting on an object you place virtually.

You need Apple A9 or above processors in order to run ARKit. So you'll find it's not compatible with older models. ARKit is optimised for use with Metal, SceneKit and even apps that make use of third-party tools such as Unreal Engine and Unity.

Metal

The Metal framework and library is used to basically make full use of your device's graphics processing unit (GPU). So if you're planning on making a graphically intensive app, you're going to need Metal.

Metal is now Metal 2 and boasts further improved performance by granting even more control on the GPU for users. Some of the improved features in Metal 2 includes:

Improved GPU efficiency

Metal 2 allows the GPU to take over even more tasks in terms of graphical processing. For example, shader resource assignment which would be normal done by the CPU will be taken over by the GPU. This improves overall efficiencies.

Metal for Mac

With Metal 2, the Metal Performance Shaders (MPS) framework becomes available for the macOS as well. This gives macOS users access to MPS' library of image processing and machine learning primitives. Additionally, MPS expands Metal's machine learning capabilities as well.

Metal 2 adds support for VR and External GPUs on the mac as well, though external GPUs aren't yet available on Mac. But we hear they're on the way.

Overall Metal 2 is an improvement over Metal and makes developing apps easier.